



On the number of perfect matchings of middle graphs[☆]

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ABSTRACT

Let G be a connected graph with n vertices and m edges, and let $L(G)$ and $M(G)$ be the line graph and middle graph of G , respectively. Let $p(G)$ be the number of perfect matchings of G . Dong, Yan and Zhang (Discrete Applied Mathematics, 161 (2013) 794–801.) proved that if m is even, then $p(L(G)) \geq 2^{m-n+1}$, and characterized the extremal graphs G with equality. In this paper, we obtain a similar result for the middle graph of G , that is, if $m + n$ is even, then $p(M(G)) \geq 2^{m-n+1} 3^{\frac{m-n}{2}}$, and characterize the extremal graphs G such that $p(M(G)) = 2^{m-n+1} 3^{\frac{m-n}{2}}$. As applications, we enumerate perfect matchings of the middle graphs of the Cartesian products $P_n \times P_m$, $C_n \times P_m$, and $C_n \times C_m$, where P_n and C_n are the path and cycle with n vertices.

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1. Introduction

In this paper, the graphs G we consider are simple. Denote by $V(G) = \{v_1, v_2, \dots, v_n\}$ and $E(G) = \{e_1, e_2, \dots, e_m\}$ the vertex set and edge set of G , respectively. For any vertex v in G , let $d_G(v)$ be the degree of vertex v , and $E_G(v)$ be the set of edges in G which are incident with v . Let $\Delta(G)$ denote the maximum degree of G . Denote by P_n and C_n the path and cycle with n vertices, respectively.

The *line graph* of G , denoted by $L(G)$, is defined as the graph with vertex set $V(L(G)) = E(G)$ such that any two vertices e and f are adjacent in $L(G)$ if and only if the two edges e and f have a common vertex in G . The *middle graph* of G , denoted by $M(G)$, is obtained from G by inserting a new vertex on each edge of G and connecting those pairs of new vertices on adjacent edges of G . Hence, $M(G)$ has vertex set $V(G) \cup E(G)$ and edge set $E(L(G)) \cup \{ue, ve | e = uv \in E(G)\}$. Fig. 1(a), (b) and (c) show a connected graph G and its corresponding line graph $L(G)$ and middle graph $M(G)$, respectively.

A *perfect matching* of a graph G is a set of independent edges of G covering all vertices of G . Perfect matchings are called *dimer configurations* in statistical physics [10] and *Kekulé structures* in quantum chemistry [3,9,12]. The dimer configurations were first posed by chemists and physicists in the 1930s. Configurations of the dimer model represent the absorption of di-atomic molecules on the surface of a crystal, and the key point of dimer problem is to how to compute the partition function, that is, to enumerate the perfect matchings of a graph. Let $p(G)$ denote the number of all perfect matchings of G . It is well known that counting perfect matchings is an NP-hard problem [4,8,14]. Note that the graph G with an odd number of vertices contains no perfect matchings, that is, $p(G) = 0$. Obviously, $p(L(G)) = 0$ if m is odd and $p(M(G)) = 0$ if $m + n$ is odd. If G is a connected claw-free graph and n is even, Sumner [11] and Vergnas [15] independently showed that $p(G) \geq 1$. Since every line graph is claw-free, $p(L(G)) \geq 1$ for every connected graph G with an even number of edges.

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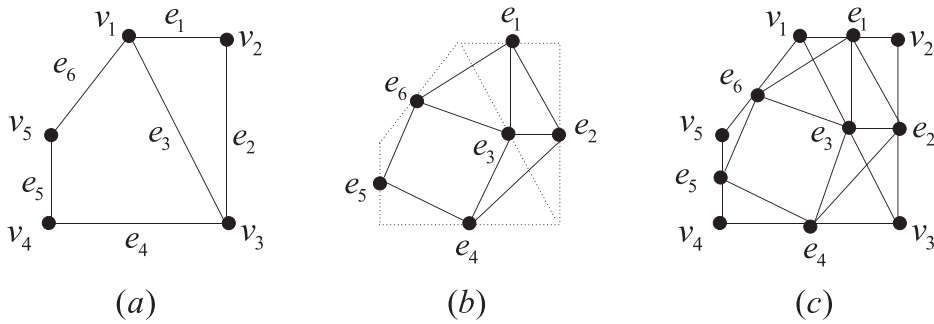


Fig. 1. (a). A connected graph G ; (b). The line graph $L(G)$ of G , which is illustrated with solid lines; (c). The middle graph $M(G)$ of G .

The *cartesian product* of graphs G and H , denoted by $G \times H$, is the graph with vertex set $V(G \times H) = \{(a, v) | a \in V(G), v \in V(H)\}$, where (a, v) is adjacent to (b, w) whenever $a = b$ and vw is an edge in H , or $v = w$ and ab is an edge in G . A seminal milestone in lattice statistics is the exact solution of the enumeration of perfect matchings on $P_n \times P_m$, $C_n \times P_m$ and $C_n \times C_m$ obtained by Kasteleyn [5] and Temperley and Fisher [13] on lattices which mimics the adsorption of diatomic molecules on a surface in 1961.

For the enumerative problem of perfect matchings on the line graphs, Ciucu, Liu, Yang [1] and Dong, Yan, Zhang [2] independently showed that if m is even and $\Delta(G) \leq 3$, then $p(L(G)) = 2^{m-n+1}$. Particularly, Dong, Yan, and Zhang [2] also showed that if m is even, then $p(L(G)) \geq 2^{m-n+1}$, and characterized the graphs G such that $p(L(G)) = 2^{m-n+1}$. For the middle graphs, Li, Li, and Yan [7] used a combinatorial technique to show that if G is a cubic graph with an even number of edges, then $p(M(G)) = 2^{\frac{n}{2}+1}3^{\frac{n}{4}}$. Recently, the current authors [6] proved that if $m + n$ is even and $\Delta(G) \leq 4$, then $p(M(G)) = 2^{m-n+1}3^{\frac{m-n}{2}}$.

In this paper, we show that if $m + n$ is even, then $p(M(G)) \geq 2^{m-n+1}3^{\frac{m-n}{2}}$, and characterize the graphs G such that $p(M(G)) = 2^{m-n+1}3^{\frac{m-n}{2}}$. As applications, we enumerate perfect matchings of the middle graphs of the Cartesian products $P_n \times P_m$, $C_n \times P_m$ and $C_n \times C_m$, respectively.

2. A lower bound for the number of perfect matchings of middle graphs

Let G be a connected graph with $V(G) = \{v_1, v_2, \dots, v_n\}$ and $E(G) = \{e_1, e_2, \dots, e_m\}$. Let G^u be the graph obtained from G by adding a new pendent edge $v_i u_i$ to each vertex v_i in G for $i = 1, 2, \dots, n$. That is, G^u is the graph with $V(G^u) = \{u_1, u_2, \dots, u_n\} \cup V(G)$ and $E(G^u) = \{v_i u_i | i = 1, 2, \dots, n\} \cup E(G)$. We call G^u to be the u -graph of G .

Proposition 2.1 ([6]). *Let G be a connected graph and G^u be the u -graph of G . Then*

$$M(G) \cong L(G^u).$$

By Proposition 2.1, $p(M(G)) = p(L(G^u))$. So for the enumerative problem of perfect matchings of middle graphs $M(G)$, we only need to enumerate $p(L(G^u))$. On the other hand, Dong, Yan, Zhang [2] gave a technique to enumerate perfect matchings of line graphs, which will be restated as follows.

For any connected graph G and any vertex v in G , let $g_G(v)$ be the number of those components of $G - v$ each of which has an even number of edges. For any non-negative integer N , let

$$(2N)!! = (2N - 1)!! = (2N - 1)(2N - 3) \cdots 3 \cdot 1,$$

and $0!! = (-1)!! = 1$. Obviously, if $N \geq 1$, then $(2N)!! = (2N - 1)!! \geq 3^{N-1}$, where the equality holds if and only if $N = 1$ or 2. Let $f_G(v) = d_G(v) - k$, where k is the number of components of $G - v$. Hence $f_G(v) \geq 0$, where the equality holds if and only if every edge of $E_G(v)$ is a bridge of G .

Proposition 2.2 ([2]). *Let T be a tree with n vertices, where n is odd and $n \geq 3$. Then*

$$p(L(T)) = \prod_{v \in V(T)} g_T(v)!!.$$

Proposition 2.3 ([2]). *Let G be a connected graph with m edges. For each vertex v in G ,*

$$f_G(v) + g_G(v) \equiv m \pmod{2}.$$

Particularly, if G is a tree and m is even, then $g_G(v)$ is even.

Let G be a connected graph with $V(G) = \{v_1, v_2, \dots, v_n\}$ and $E(G) = \{e_1, e_2, \dots, e_m\}$, where $m - n + 1 > 0$. Suppose that T^* is a spanning tree of G and $E(G) \setminus E(T^*) = \{e_1, e_2, \dots, e_{m-n+1}\}$. Let v_{s_i} and v_{t_i} be the two ends of e_i for $i = 1, 2, \dots, m - n + 1$. For any $(m - n + 1)$ -tuple $(j_1, j_2, \dots, j_{m-n+1})$, where $j_i \in \{0, 1\}$ and $1 \leq i \leq m - n + 1$, let $G(j_1, j_2, \dots, j_{m-n+1})$ be the graph obtained from T^* by adding $(m - n + 1)$ new vertices $w_1, w_2, \dots, w_{m-n+1}$ and adding a new edge joining w_i to v_{s_i} if $j_i = 0$ else to v_{t_i} if $j_i = 1$ for every $i = 1, 2, \dots, m - n + 1$. Note that $G(j_1, j_2, \dots, j_{m-n+1})$ is a tree with $m + 1$ vertices and m edges, and $w_1, w_2, \dots, w_{m-n+1}$ are pendent vertices of $G(j_1, j_2, \dots, j_{m-n+1})$. Set

$$H_{T^*}(G) = \{G(j_1, j_2, \dots, j_{m-n+1}) | j_i \in \{0, 1\}, 1 \leq i \leq m - n + 1\}.$$

Hence $|H_{T^*}(G)| = 2^{m-n+1}$.

Proposition 2.4 ([2]). *Let G be a connected graph and T^* be a spanning tree of G . Then*

$$p(L(G)) = \sum_{T \in H_{T^*}(G)} p(L(T)).$$

Now, let T be a tree with $V(T) = \{v_1, v_2, \dots, v_n\}$ and $E(T) = \{e_1, e_2, \dots, e_{n-1}\}$. For each edge $e_i \in E(T)$, let $T - e_i = A_{i1} \cup A_{i2}$, where A_{i1}, A_{i2} are the two components of $T - e_i$. Let

$$U_E = \{A_{i1}, A_{i2} | i = 1, 2, \dots, n - 1\}.$$

Hence $|U_E| = 2n - 2$. For each vertex $v_i \in V(T)$, let $T - v_i = B_{i1} \cup B_{i2} \cup \dots \cup B_{id_i}$, where $B_{i1}, B_{i2}, \dots, B_{id_i}$ are the d_i components of $T - v_i$. Obviously, $d_i = d_T(v_i)$. Let

$$U_V = \{B_{i1}, B_{i2}, \dots, B_{id_i} | i = 1, 2, \dots, n\}.$$

Hence $|U_V| = \sum_{i=1}^n d_i = 2n - 2$.

Theorem 2.5. *Let T be a tree with $V(T) = \{v_1, v_2, \dots, v_n\}$ and $E(T) = \{e_1, e_2, \dots, e_{n-1}\}$, and let U_E and U_V be the sets defined as above. Then $U_E = U_V$.*

Proof. Let A_{ij} be an arbitrary element in U_E . Without loss of generality, suppose that A_{ij} is a component of $T - e_i$ and $e_i = v_i v_{i+1}$, where $v_{i+1} \in V(A_{ij})$. It is obvious that A_{ij} is also a component of $T - v_i$. That is, $A_{ij} \in U_V$. Hence $U_E \subseteq U_V$.

Let B_{ij} be an arbitrary element in U_V . Without loss of generality, suppose that B_{ij} is a component of $T - v_i$ and v_{i+1} is the unique vertex of B_{ij} which is incident with v_i . Let $e_i = v_i v_{i+1}$. Obviously, B_{ij} is also a component of $T - e_i$. That is, $B_{ij} \in U_E$. Hence $U_V \subseteq U_E$.

Therefore, $U_E = U_V$. The theorem thus follows. $\square$

Corollary 2.6. *Let T be a tree with $V(T) = \{v_1, v_2, \dots, v_n\}$ and $E(T) = \{e_1, e_2, \dots, e_{n-1}\}$, where n is odd. Then*

$$\sum_{i=1}^n g_T(v_i) = n - 1.$$

Proof. For each edge $e_i \in E(T)$, let $T - e_i = A_{i1} \cup A_{i2}$. Obviously, $|E(A_{i1})|$ and $|E(A_{i2})|$ have different parity since $|E(T)|$ is even. Without loss of generality, suppose that $|E(A_{i1})|$ is even and $|E(A_{i2})|$ is odd for $i = 1, 2, \dots, n - 1$. By Theorem 2.5, we have

$$U_V = U_E = \{A_{i1} | i = 1, 2, \dots, n - 1\} \cup \{A_{i2} | i = 1, 2, \dots, n - 1\}.$$

This implies that $\sum_{i=1}^n g_T(v_i) = n - 1$. The corollary thus follows. $\square$

Theorem 2.7. *Let G be a connected graph with $V(G) = \{v_1, v_2, \dots, v_n\}$ and $E(G) = \{e_1, e_2, \dots, e_m\}$, where $m + n$ is even. Then*

$$p(M(G)) \geq 2^{m-n+1} 3^{\frac{m-n}{2}}.$$

Proof. Let T^* be a spanning tree of G and let T^{*u} be the u -graph of T^* , which has vertex set $V(T^{*u}) = \{u_1, u_2, \dots, u_n\} \cup V(T^*)$. Obviously, T^{*u} is a spanning tree of G^u , where G^u is the u -graph of G . By the definition of the set $H_{T^{*u}}(G^u)$ (the upper paragraph of Proposition 2.4), it is obvious that $|H_{T^{*u}}(G^u)| = 2^{m-n+1}$. By Propositions 2.1 and 2.4,

$$p(M(G)) = p(L(G^u)) = \sum_{T \in H_{T^{*u}}(G^u)} p(L(T)), \quad (2.1)$$

where $V(T) = V(T^{*u}) \cup \{w_1, w_2, \dots, w_{m-n+1}\}$. For every tree $T \in H_{T^{*u}}(G^u)$, $|E(T)| = m + n$ is even and $d_T(u_i) = d_T(w_j) = 1$. Hence $g_T(u_i)!! = g_T(w_j)!! = 0!! = 1$, where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m - n + 1$. By Proposition 2.2,

$$p(L(T)) = \prod_{v \in V(T)} g_T(v)!! = \prod_{i=1}^n g_T(v_i)!!, \quad (2.2)$$

and by [Corollary 2.6](#),

$$\sum_{i=1}^n g_T(v_i) = \sum_{v \in V(T)} g_T(v) = m + n. \quad (2.3)$$

For every vertex v_i in T , since $|E(T)| = m + n$ is even and $\{u_i\}$ is a component of $T - v_i$, $g_T(v_i) \geq 2$ is even by [Proposition 2.3](#). Hence

$$g_T(v_i)!! \geq 3^{\frac{g_T(v_i)}{2}-1}. \quad (2.4)$$

By Eqs. (2.2)–(2.4), we can get

$$p(L(T)) \geq \prod_{i=1}^n 3^{\frac{g_T(v_i)}{2}-1} = 3^{\sum_{i=1}^n (\frac{g_T(v_i)}{2}-1)} = 3^{\frac{m-n}{2}}. \quad (2.5)$$

By Eqs. (2.1) and (2.5), $p(M(G)) \geq 2^{m-n+1} 3^{\frac{m-n}{2}}$. The theorem thus follows. $\square$

Remark 2.8. From the proof of [Theorem 2.7](#), $p(L(T)) \geq 3^{\frac{m-n}{2}}$ for every tree $T \in H_{T^*u}(G^u)$, where the equality holds if and only if $g_T(v_i) = 2$ or 4 for every $i = 1, 2, \dots, n$.

3. The extremal graphs G with $p(M(G)) = 2^{m-n+1} 3^{\frac{m-n}{2}}$

For any connected graph G and any vertex v in G , let $h_G(v)$ be the number of those components H of $G - v$ each of which has an odd cyclomatic number (i.e., $|E(H)| - |V(H)| + 1$ is odd).

Lemma 3.1. Let G be a connected graph with $V(G) = \{v_1, v_2, \dots, v_n\}$ and G^u be the u -graph of G . Then for every vertex $v_i \in V(G)$,

$$f_{G^u}(v_i) = f_G(v_i), \quad g_{G^u}(v_i) = h_G(v_i) + 1.$$

Proof. It is obvious that $f_{G^u}(v_i) = f_G(v_i)$. Now let $G - v_i = G_1 \cup G_2 \cup \dots \cup G_k$, where $G_1, G_2, \dots, G_k$ are the k components of $G - v_i$. Then $G^u - v_i = G_1^u \cup G_2^u \cup \dots \cup G_k^u \cup \{u_i\}$, where G_t^u is the u -graph of G_t for $1 \leq t \leq k$. Note that $|E(G_t^u)| = |E(G_t)| + |V(G_t)|$ and $c(G_t) = |E(G_t)| - |V(G_t)| + 1$, where $c(G_t)$ is the cyclomatic number of G_t . Hence $|E(G_t^u)|$ and $c(G_t)$ have different parity. That is, if G_t^u is a component of G^u which has an even (odd) number edges, then G_t is the component of G which has an odd (even) cyclomatic number for $i = 1, 2, \dots, k$. Obviously, $E(\{u_i\}) = 0$. Hence $g_{G^u}(v_i) = h_G(v_i) + 1$. The lemma thus follows. $\square$

Proposition 3.2 ([2]). Let G be a connected graph and e be an edge with ends u and v . If e is on some cycle of G , then for every vertex $x \in V(G)$,

$$g_{G(u,w)}(x) + f_{G(u,w)}(x) \leq g_G(x) + f_G(x); \quad g_{G(v,w)}(x) + f_{G(v,w)}(x) \leq g_G(x) + f_G(x).$$

Corollary 3.3. Let G be a connected graph and T^* be a spanning tree of G . Then for every tree $T \in H_{T^*}(G)$ and every vertex $v \in V(G)$,

$$g_T(v) \leq g_G(v) + f_G(v).$$

Proof. By the definition of $H_{T^*}(G)$ (see the definition in the last paragraph before [Proposition 2.4](#)), and applying [Proposition 3.2](#) repeatedly, the result follows immediately. $\square$

On the other hand, Zhou [16] also gave an important proposition.

Proposition 3.4 ([16]). Let G be a connected graph and T^* be a spanning tree of G . For each $v \in V(G)$, then there exists a tree $T \in H_{T^*}(G)$ such that $g_T(v) = f_G(v) + g_G(v)$.

Theorem 3.5. Let G be a connected graph with $V(G) = \{v_1, v_2, \dots, v_n\}$ and $E(G) = \{e_1, e_2, \dots, e_m\}$, where $m + n$ is even. Then $p(M(G)) = 2^{m-n+1} 3^{\frac{m-n}{2}}$ if and only if $f_G(v_i) + h_G(v_i) \in \{1, 3\}$ for every vertex $v_i \in V(G)$.

Proof. Let G^u be the u -graph of G and $V(G^u) = \{v_1, v_2, \dots, v_n\} \cup \{u_1, u_2, \dots, u_n\}$. By [Lemma 3.1](#), $f_{G^u}(v_i) + g_{G^u}(v_i) = f_G(v_i) + h_G(v_i) + 1$. Note that $|E(G^u)| = m + n$ is even. Hence, by [Proposition 2.3](#), $f_G(v_i) + h_G(v_i)$ is odd for $i = 1, 2, \dots, n$. So, $f_G(v_i) + h_G(v_i) \in \{1, 3\}$ if and only if $f_G(v_i) + h_G(v_i) \leq 3$.

We prove that the necessity holds by contradiction. Suppose that $f_G(v_i) + h_G(v_i) \geq 5$ for some vertex v_i in G . That is, $f_{G^u}(v_i) + g_{G^u}(v_i) \geq 6$. By Proposition 3.4, there exists a tree $T \in H_{T^{*u}}(G^u)$ such that $g_T(v_i) = f_{G^u}(v_i) + g_{G^u}(v_i) \geq 6$, where T^{*u} is a spanning tree of G^u . By Remark 2.8, $p(L(T)) > 3^{\frac{m-n}{2}}$. By Eq. (2.1), we can get

$$p(M(G)) = p(L(G^u)) = \sum_{T \in H_{T^{*u}}(G^u)} p(L(T)) > 2^{m-n+1} 3^{\frac{m-n}{2}},$$

contradicting $p(M(G)) = 2^{m-n+1} 3^{\frac{m-n}{2}}$. Hence $f_G(v_i) + h_G(v_i) \in \{1, 3\}$ for every v_i in G .

For the sufficiency, we suppose that $p(M(G)) > 2^{m-n+1} 3^{\frac{m-n}{2}}$. Then the proof of Theorem 2.7 implies that there exists a tree $T \in H_{T^{*u}}(G^u)$ such that $p(L(T)) > 3^{\frac{m-n}{2}}$. By Remark 2.8, there must be some vertex v_i satisfying $g_T(v_i) \geq 5$. By Corollary 3.3, $g_T(v_i) \leq f_{G^u}(v_i) + g_{G^u}(v_i) = f_G(v_i) + h_G(v_i) + 1$. That is, $f_G(v_i) + h_G(v_i) \geq 4$, contradicting $f_G(v_i) + h_G(v_i) \in \{1, 3\}$ for every v_i in G . Hence $p(M(G)) = 2^{m-n+1} 3^{\frac{m-n}{2}}$. The theorem thus follows. $\square$

4. Applications

As applications of Theorem 3.5, in this section, we enumerate perfect matchings of the middle graphs for $P_n \times P_m$, $C_n \times P_m$ and $C_n \times C_m$, respectively.

For any connected graph G and any vertex v of G , it is obvious that $f_G(v) + h_G(v) \leq d_G(v)$. By Theorem 3.5, the following corollary follows immediately.

Corollary 4.1 ([6]). *Let G be a connected graph with n vertices and m edges, where $m+n$ is even. If the maximum degree $\Delta(G)$ of G satisfies $\Delta(G) \leq 4$, then*

$$p(M(G)) = 2^{m-n+1} 3^{\frac{m-n}{2}}.$$

Theorem 4.2. *For the Cartesian products $P_n \times P_m$, $C_n \times P_m$ and $C_n \times C_m$,*

- (1) $p(M(P_n \times P_m)) = 2^{mn-(m+n)+1} 3^{\frac{mn-(m+n)}{2}}$, if both n and m are even;
- (2) $p(M(C_n \times P_m)) = 2^{mn-n+1} 3^{\frac{mn-n}{2}}$, if either n is even or both m and n are odd;
- (3) $p(M(C_n \times C_m)) = 2^{mn+1} 3^{\frac{mn}{2}}$, if mn is even.

Proof. Note that $P_n \times P_m$, $C_n \times P_m$ and $C_n \times C_m$ are actually square lattice graphs on the plane, cylinder and sphere, respectively. Hence $\Delta(P_n \times P_m) \leq 4$, $\Delta(C_n \times P_m) \leq 4$ and $\Delta(C_n \times C_m) \leq 4$. By Corollary 4.1, it suffices to prove that the sum of the number of vertices and the number of edges is even for graphs $M(P_n \times P_m)$, $M(C_n \times P_m)$ and $M(C_n \times C_m)$, respectively.

(1) If both n and m are even, then $|V(P_n \times P_m)| + |E(P_n \times P_m)| = 3mn - (m+n)$ is even since $|V(P_n \times P_m)| = mn$ and $|E(P_n \times P_m)| = (m-1)n + (n-1)m = 2mn - (m+n)$.

(2) If either n is even or both m and n are odd, then $|V(C_n \times P_m)| + |E(C_n \times P_m)| = 3mn - n$ is even since $|V(C_n \times P_m)| = mn$ and $|E(C_n \times P_m)| = (m-1)n + mn = 2mn - n$.

(3) If mn is even, then $|V(C_n \times C_m)| + |E(C_n \times C_m)| = 3mn$ is even since $|V(C_n \times C_m)| = mn$ and $|E(C_n \times C_m)| = mn + nm = 2mn$.

The theorem thus follows. $\square$

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Data availability

No data was used for the research described in the article.

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